A BOUND OF 212 FOR GAPS BETWEEN CONSECUTIVE PRIMES

OPENAI GPT-5.6 SOL

ABSTRACT. We prove that every admissible 45-tuple has infinitely many translates containing at least two primes. In particular, $\liminf_{n\to\infty}(p_{n+1}-p_n) \le 212$. The distributional input is Stadlmann's equidistribution estimates for moduli with a large smooth factor. A three-part partition lemma enlarges the permissible support, and a sum of three product weights with separate radial profiles supplies the sieve function. Its integrals are reduced to finite convolutions and prefix sums. A directed-rounding certificate gives $45J_{\mathrm{box}}/I > 1.0001949699$; the box numerator is a lower bound for the required sieve integral. The exact coefficients and a separately implemented verifier are included in the ancillary material and at `https://github.com/Siddhartha-Mahajan/prime-gap-212`.

1. INTRODUCTION

Let p_n denote the nth prime, and put

$$H_1 = \liminf_{n\to\infty}(p_{n+1}-p_n).$$

The sieve of Goldston, Pintz and Yıldırım [2] provided the starting point for the modern theory of small prime gaps. Zhang's theorem [8] established that H_1 is finite. Maynard's multidimensional sieve [3], together with Tao's independent work and the subsequent Polymath project [5], led to $H_1 \le 246$. Stadlmann [7] obtained $H_1 \le 240$ by combining the epsilon-enlargement of the sieve support with equidistribution estimates for moduli having a large smooth factor. These estimates allow some nonsmooth factors to remain, provided that they can be allocated to appropriate divisor ranges.

We prove the following result.

Theorem 1.1. *One has $H_1 \le 212$.*

A finite set of integers is *admissible* if it omits a residue class modulo every prime. We obtain Theorem 1.1 from the stronger statement below.

Theorem 1.2. *Let $\{h_1,\dots,h_{45}\}$ be an admissible set of distinct integers. There are infinitely many integers n for which at least two of $n+h_1,\dots,n+h_{45}$ are prime.*

There are two ingredients beyond the sieve and distribution estimates of [7]. The first is a partition argument for the logarithms of the nonsmooth factors. The support is described by four bounds: one for a single large coordinate, one for two, one for three, and one for four or more. We retain the dependence between the available divisor capacities, rather than replacing each capacity by its separate minimum. This gives a uniform three-part partition in the most restrictive Type II range. A direct factorization argument retains a positive auxiliary slack throughout this range.

Date: September 2, 2026.

This is a substantially AI-generated mathematical manuscript. Siddhartha Mahajan, Research Fellow at Lossfunk, supplied the problem and source materials, gave brief high-level directions, and arranged its release. He did not independently derive or verify the mathematical argument. The research and manuscript were generated in the ChatGPT web interface by GPT-5.6 Sol using Pro reasoning effort; Codex was used afterward for review and computational verification. See Appendix B for the full provenance and release status. The public repository is `https://github.com/Siddhartha-Mahajan/prime-gap-212`. Human correspondence: Siddhartha Mahajan, `siddharthamahajan03@gmail.com`.

The second ingredient is the choice and evaluation of the sieve function. We use three product weights, each multiplied by a function of the sum of the coordinates. After restricting this function to a union of boxes, both quadratic forms can be expressed through one-dimensional convolutions. The function on this union is a finite, exactly specified object. In particular, the proof does not require an error estimate for approximating an unspecified continuous optimizer, nor does it require certifying an eigenvalue calculation.

The distribution theorems themselves are not strengthened here. We use Lemmas 3, 4, 6, and 7 of [7], which extend estimates from [1, 4, 6]. The role of the new partition lemma is to place a larger family of moduli within their hypotheses. We use a nonnegative prime-supported weight of asymptotic density one, so that no loss from a prime minorant enters the variational inequality. A direct Heath–Brown reduction at the endpoint $\xi_2 = \xi_3 = 2/5$ supplies the required prime-distribution estimate.

Sections 2–5 establish the permissible support. Section 6 gives the finite quadratic forms. Section 7 records the certificate and proves the two theorems. The finite certificate is specified in Appendix A, and the generation history and release status are recorded in Appendix B.

1.1. **Notation.** All integrals are Lebesgue integrals. A list may be empty; its sum is then zero. The entries in the lists used for factorization are real numbers. We write $P(x) = \prod_{p<x} p$, and say that an integer is x^δ-smooth if all of its prime factors are less than x^δ. All moduli to which we apply a distribution estimate are squarefree. The parameters ε, η, ε_0, and τ have different roles: ε enlarges the sieve support, $\eta = 10^{-10}$ is the tolerance in the convolution decomposition, ε_0 contracts the support when sieve weights are constructed, and $\tau > 0$ is the auxiliary parameter in the distribution estimates.

2. The sieve criterion

Let $k \geq 2$, and suppose that

$$0 < \varepsilon < A < \frac{1}{2} - \varepsilon, \qquad 0 < \delta < B_m \leq B_{m+1} \leq B_m + \delta \quad (m \geq 1), \quad B_1 < 1. \tag{2.1}$$

For $\mathbf{t} = (t_1, \ldots, t_k) \in [0, \infty)^k$, define

$$r(\mathbf{t}) = \#\{i : t_i > \delta\}.$$

We use the support

$$\mathcal{T}_k = \left\{ \mathbf{t} \in [0, \infty)^k : \sum_i t_i < A + \varepsilon, \quad \sum_{i : t_i > \delta} t_i \leq B_{r(\mathbf{t})} \right\}, \tag{2.2}$$

where the second condition is omitted when $r(\mathbf{t}) = 0$. This is the one-layer case of [7, Definition 1], with $\underline{A} = (-\varepsilon, A)$.

A useful consequence of (2.1) is the following inheritance property. If a list of r numbers, all at least δ, has sum at most B_r, then every sublist of j numbers has sum at most B_j. Indeed, deleting $r - j$ entries decreases the sum by at least $(r - j)\delta$, whereas

$$B_r \leq B_j + (r - j)\delta. \tag{2.3}$$

In particular, each coordinate of (2.2) is at most B_1, apart from immaterial boundary conventions.

For a symmetric real function $F \in L^2(\mathcal{T}_k)$, extended by zero, set

$$I(F) = \int_{[0,\infty)^k} F(\mathbf{t})^2 \, \mathrm{d}\mathbf{t}, \tag{2.4}$$

$$J(F) = \int_{\substack{\mathbf{t} \in [0,\infty)^{k-1} \\ t_1 + \cdots + t_{k-1} \leq A - \varepsilon}} \left(\int_0^\infty F(\mathbf{t}, u) \, \mathrm{d}u \right)^2 \mathrm{d}\mathbf{t}. \tag{2.5}$$

We specify the required family of moduli. For $0 < \varepsilon_0 < 1$, let $\mathcal{Q}(x;\varepsilon_0)$ consist of the squarefree integers admitting a factorization

$$q = ee' \prod_{i=1}^{m} f_i \prod_{i=1}^{m'} f_i' \tag{2.6}$$

such that ee' is x^δ-smooth, $\log_x f_i, \log_x f_i' \geq \delta$, and

$$\log_x \prod_i f_i \leq (1-\varepsilon_0)B_m, \qquad \log_x \prod_i f_i' \leq (1-\varepsilon_0)B_{m'}, \tag{2.7}$$

$$\log_x \left(e \prod_i f_i \right) \leq (1-\varepsilon_0)(A-\varepsilon), \qquad \log_x \left(e' \prod_i f_i' \right) \leq (1-\varepsilon_0)(A+\varepsilon). \tag{2.8}$$

A condition on an empty product in (2.7) is omitted. Thus

$$q \leq x^{2A(1-\varepsilon_0)}. \tag{2.9}$$

This is the squarefree part of the one-layer family in [7, Definition 2].

For a function $f(n;x)$ write

$$E_f(x;q,a) = \sum_{\substack{x \leq n \leq 2x \\ n \equiv a \pmod q}} f(n;x) - \frac{1}{\phi(q)} \sum_{\substack{x \leq n \leq 2x \\ (n,q)=1}} f(n;x).$$

For positive integers n define the nonnegative prime-supported weight

$$\rho_x(n) = \frac{\log n}{\log(3x)} \mathbf{1}_{\mathbb{P}}(n) \mathbf{1}_{x/2 \leq n \leq 3x}. \tag{2.10}$$

Thus $0 \leq \rho_x \leq \mathbf{1}_{\mathbb{P}}$ on the range relevant to the sieve, and the prime number theorem gives

$$\sum_{x \leq n \leq 2x} \rho_x(n) = (1+o(1)) \frac{x}{\log x}.$$

The distribution statement needed below is

$$\sum_{q \in \mathcal{Q}(x;\varepsilon_0)} |E_{\rho_x}(x;q,a)| \ll_{C,\varepsilon_0} \frac{x}{(\log x)^C} \qquad \text{for every } C > 0, \tag{2.11}$$

with the residue convention $(a, P(x)) = 1$ of [7, Definition 3].

Proposition 2.1 (One-layer sieve criterion)**.** *Suppose* (2.1) *and* (2.11) *hold. If a symmetric real* $F \in L^2(\mathcal{T}_k)$ *satisfies*

$$kJ(F) > I(F) > 0, \tag{2.12}$$

then every admissible k-tuple has infinitely many translates containing at least two primes.

Proof. Apply [7, Proposition 1 and its proof] with one support layer and $\rho = \rho_x$. The minorant parameters are $c_1 = c_2 = 0$. The roughness condition is automatic because ρ_x is supported on primes, and its mean was recorded above. In the one-layer case the integrals in that proposition are exactly (2.4) and (2.5). The sieve argument applies to any fixed admissible tuple.

The stated L^2 formulation includes nonsmooth functions. One can also see the relevant continuity directly: integration in the last coordinate has squared operator norm at most B_1, by Cauchy–Schwarz. Hence J is continuous on $L^2(\mathcal{T}_k)$. The contraction and smooth approximation in [7, Lemma 1] therefore preserve the strict inequality (2.12). □

3. Partition lemmas

The next two lemmas concern two lists of positive real numbers. A j-subsum means the sum of any j distinct entries from one list, not necessarily consecutive entries.

Lemma 3.1. *Let $0 < b_1 \le b_2 \le b_3$, and suppose that every j-subsum of either of two lists is at most b_j, for $1 \le j \le 3$. Let $a, c > 0$ satisfy*

$$2b_1 \le a, \qquad \frac{5}{2}b_2 \le a + c, \qquad b_3 \le 3c. \tag{3.1}$$

The sum of all entries exceeding c is at most a. Moreover, if the combined total is M and $2(M - a) \le c$, the entries can be partitioned into parts of sums at most a and c.

Proof. There are at most two entries exceeding c in each list, since three such entries would have sum greater than b_3. If each list has at most one, their combined sum is at most $2b_1 \le a$. Otherwise one list has two entries exceeding c, so $c < b_2/2$. The sum of all entries exceeding c is then at most $2b_2$, and

$$a \ge \frac{5}{2}b_2 - c > 2b_2.$$

This proves the first assertion.

If $M \le a$, there is nothing to prove. Otherwise put $R = M - a > 0$. An entry in $[R, c]$ can be placed in the second part on its own. If no such entry exists, every entry at most c is less than R. The first assertion implies that these smaller entries have total at least R. Accumulate them until their sum first reaches R. The resulting sum is less than $2R \le c$, and the sum of the remaining entries is at most a. □

Lemma 3.2 (Three-part partition)**.** *Let $0 < \delta < b_1 \le b_2 \le b_3 \le B$. Suppose that all entries in two lists are at least δ, each list has total at most B, and every j-subsum of either list is at most b_j, for $1 \le j \le 3$. Let $a, c, d > 0$, and put $c' = \max(c, d)$, $d' = \min(c, d)$. Suppose that*

$$2b_1 \le a, \qquad \frac{5}{2}b_2 \le a + c', \qquad b_3 \le 3c', \tag{3.2}$$

$$2B \le a + d', \qquad 4B \le 2a + c' + 2\delta. \tag{3.3}$$

Then the entries can be partitioned into three parts of sums at most a, c, d, respectively.

Proof. Let M be the combined total. We may assume that $M > a$. Suppose first that there is an entry $y \le d'$. Place y in the part of capacity d'. Deleting this entry preserves all the subsum conditions. The remaining total $M' = M - y$ satisfies

$$2(M' - a) \le 2(2B - \delta - a) \le c'.$$

Lemma 3.1 partitions the remaining entries into the parts of capacities a and c'.

If there is no entry at most d', then every entry exceeds d'. By (3.3), $R = M - a \le 2B - a \le d'$. By the first assertion of Lemma 3.1, the entries exceeding c' have total at most a. Since $M > a$, some entry is at most c'. This entry is greater than $d' \ge R$, so placing it in the part of capacity c' leaves total at most a. The part of capacity d' is empty. □

For completeness, we record the elementary fact used to turn a partition of the large factors into a factorization of the modulus.

Lemma 3.3 (Filling an interval)**.** *Let $z_1, \ldots, z_s \in (0, \delta)$, and let $Y \ge 0$. If $U - L \ge \delta$, $Y \le U$, and $Y + \sum_i z_i \ge L$, some sublist has sum Z with $Y + Z \in [L, U]$. If instead $Y < U$ and $Y + \sum_i z_i > L$, one can obtain $Y + Z \in (L, U)$ whenever $U - L > \delta$.*

Proof. If the initial value is already in the desired interval, use the empty sublist. Otherwise add entries until the lower endpoint is first reached, or strictly exceeded in the open-interval version. The last increment is less than δ. □

4. The critical divisor factorization

We now fix the parameters used in the proof. All entries in the following display are exact:

$$\begin{gathered} k=45, \qquad A=\frac{16862}{65536}, \quad \varepsilon=\frac{557}{65536}, \quad \delta=\frac{1019}{65536}, \\ b_1=\frac{10171}{65536}, \quad b_2=\frac{10378}{65536}, \quad b_3=\frac{11321}{65536}, \quad B=\frac{12082}{65536}, \\ B_1=b_1, \qquad B_2=b_2, \qquad B_3=b_3, \qquad B_m=B \quad (m\geq 4), \\ \xi_1=\frac{19}{50}, \qquad \xi_2=\xi_3=\frac{2}{5}, \qquad \eta=10^{-10}, \quad \tau_0=10^{-8}, \quad \zeta=10^{-6}. \end{gathered} \tag{4.1}$$

These parameters satisfy (2.1). Put

$$\begin{gathered} \omega=A-\frac{1}{4}=\frac{239}{32768}, \qquad G=\frac{1}{3}+8\omega+\frac{7}{3}\delta=\frac{28047}{65536}, \\ \gamma_-=\xi_2-\eta, \qquad \gamma_+=G+3\eta. \end{gathered} \tag{4.2}$$

We will take $\tau>0$ sufficiently small, always with $\tau\leq\tau_0$. Define

$$\alpha=\gamma_--2\delta-8\omega-109\tau_0, \tag{4.3}$$

$$S=\frac{1}{2}-2\delta-10\omega-116\tau_0, \tag{4.4}$$

$$C_*=\max\left(\frac{1}{2}-\gamma_+-2\omega-7\tau_0, \frac{4}{5}\left(\frac{1}{2}-\gamma_+\right)\right). \tag{4.5}$$

Direct rational comparison gives

$$2b_1<\alpha, \qquad \frac{5}{2}b_2<S, \qquad b_3<3C_*, \tag{4.6}$$

$$2B<S, \qquad 2B<\gamma_--2\delta-4\tau_0, \qquad 4B<S+\alpha+2\delta. \tag{4.7}$$

Quantitative margins are listed in Appendix A.

Proposition 4.1. *Let $q\in\mathcal{Q}(x;\varepsilon_0)$ satisfy $q\geq x^{1/2-2\tau}$. For every $\gamma\in[\gamma_-,\gamma_+]$, there are divisors $r\mid q$, $u\mid q/r$, and $d_1\mid r$ such that, with $\delta_*=\delta+\zeta$,*

$$x^{\gamma-3\tau-\delta_*}<r<x^{\gamma-3\tau}, \tag{4.8}$$

$$x^{1-\gamma-6\tau-\delta_*}q^{-1}<u<x^{1-\gamma-6\tau}q^{-1}, \tag{4.9}$$

$$r^2x^{2-\gamma-52\tau-\delta_*}q^{-4}<d_1<r^2x^{2-\gamma-52\tau}q^{-4}. \tag{4.10}$$

Proof. Write $q=x^{1/2+2\omega_0}$ exactly. By (2.9), $-\tau\leq\omega_0\leq\omega$. Apply the inheritance property (2.3) to the two lists of logarithms in (2.6). Each list has total at most B, and its j-subsums are bounded by b_j for $1\leq j\leq 3$.

Set

$$a=\gamma-2\delta-8\omega_0-109\tau, \qquad c=\frac{1}{2}-\gamma-2\omega_0-7\tau, \qquad d=8\omega_0+105\tau. \tag{4.11}$$

These numbers are positive; in particular $d\geq 97\tau$. Writing $c'=\max(c,d)$ and $d'=\min(c,d)$, we have

$$a\geq\alpha, \qquad a+c\geq S, \qquad a+d=\gamma-2\delta-4\tau\geq\gamma_--2\delta-4\tau_0, \tag{4.12}$$

$$c'\geq c\geq\frac{1}{2}-\gamma_+-2\omega-7\tau_0, \qquad c'\geq\frac{4c+d}{5}=\frac{4}{5}\left(\frac{1}{2}-\gamma\right)+\frac{77}{5}\tau. \tag{4.13}$$

Thus $c'\geq C_*$. Moreover

$$a+d'=\min(a+c,a+d), \qquad 2a+c'\geq a+(a+c)\geq\alpha+S.$$

Equations (4.6)–(4.7) verify all hypotheses of Lemma 3.2. We obtain three parts of the large logarithms with respective sums

$$Y_1 \le a, \qquad Y_2 \le c, \qquad Y_4 \le d. \tag{4.14}$$

The indices $1, 2, 4$ identify their roles in the factorization; no fourth nonempty part is used.

Write the prime-factor logarithms of the smooth part ee' as $z_1, \ldots, z_s \in (0, \delta)$, and put

$$\begin{aligned} \ell_1 &= \gamma - 4\tau - \delta, & u_1 &= \gamma - 4\tau, \\ \ell_2 &= \frac{1}{2} - \gamma - 2\omega_0 - 7\tau - \delta, & u_2 &= c, \\ \ell_3 &= -\gamma - 8\omega_0 - 101\tau - \delta, & u_3 &= -\gamma - 8\omega_0 - 101\tau. \end{aligned}$$

Each interval has length δ, and

$$2\ell_1 + u_3 = a, \qquad \ell_1 - 2u_1 - \ell_3 = d, \qquad a + d = \gamma - 2\delta - 4\tau. \tag{4.15}$$

First construct u from the large factors in the second part and some smooth primes. The initial logarithm is at most u_2, and the total available logarithm exceeds ℓ_2, since

$$\frac{1}{2} + 2\omega_0 - (a + d) - \ell_2 = 4\omega_0 + 3\delta + 11\tau \ge 3\delta + 7\tau > 0.$$

Lemma 3.3 gives $\log_x u \in [\ell_2, u_2]$. Next construct r from the large factors in the first and third parts and some of the remaining smooth primes. Their initial logarithm is at most $a + d = u_1 - 2\delta$, and the total available logarithm is at least

$$\frac{1}{2} + 2\omega_0 - u_2 = \gamma + 4\omega_0 + 7\tau \ge \gamma + 3\tau > \ell_1.$$

Thus $v = \log_x r \in [\ell_1, u_1]$ can be obtained.

Finally, use the large factors in the first part and smooth primes already selected for r. The initial logarithm is at most $a = 2\ell_1 + u_3 \le 2v + u_3$, while the total available logarithm is at least

$$v - Y_4 \ge \ell_1 - d = 2u_1 + \ell_3 \ge 2v + \ell_3.$$

A further application of Lemma 3.3 gives a divisor $d_1 \mid r$ with

$$2v + \ell_3 \le \log_x d_1 \le 2v + u_3.$$

All selections use disjoint prime factors where required, since q is squarefree.

The closed intervals used for r and u lie strictly inside (4.8) and (4.9); their lower-endpoint slack is $\zeta - \tau > 0$ and their upper-endpoint slack is $\tau > 0$. For the last divisor, write

$$T = 2\log_x r + 2 - \gamma - 4\log_x q.$$

The constructed interval is

$$[T - 101\tau - \delta, T - 101\tau].$$

It lies strictly inside (4.10), with lower and upper slacks $\zeta - 49\tau > 0$ and $49\tau > 0$. This proves the proposition. $\square$

5. Verification of the distribution hypotheses

We use the coefficient-sequence and Siegel–Walfisz conventions of [7, Definitions 6–9]. We shall only need the Type II and Type III members of the convolution family in Definition 9 of that paper. Its Type II sequences have two Siegel–Walfisz factors, one at scale x^γ with $\xi_2 - \eta \le \gamma \le 1 - \xi_2 + \eta$. The Type III scales satisfy the conditions in that definition. We retain all of its coefficient bounds and uniformity conventions.

The following reduction derives the required prime-supported estimate from the Type II and Type III distribution estimates.

Lemma 5.1 (Endpoint Heath–Brown reduction)**.** *Let $\mathcal{R}(x)$ be a family of squarefree moduli satisfying*

$$x^{1/2-2\tau} \le q \le x^{1/2+2\omega} \qquad (q \in \mathcal{R}(x)).$$

Suppose that, for every primitive residue class a, with arbitrary logarithmic saving and uniformly in the coefficient sequences, the discrepancy sum over $\mathcal{R}(x)$ is bounded by $x(\log x)^{-C}$ for

(i) *every Type II convolution in $\mathcal{H}(\xi_1, 2/5, 2/5)$, and*
(ii) *every Type III convolution in $\mathcal{H}(\xi_1, 2/5, 2/5)$.*

If $\omega < 1/20$ and $0 < \tau \le 10^{-8}$, then the same conclusion holds for ρ_x.

Proof. We adapt the proof of [4, Lemma 2.7 and Section 3], retaining only the alternatives that are needed at the endpoint 2/5. We first record the uniformity in that argument. Fix the saving $C > 0$ required in the conclusion. Choose a fixed localization parameter $C_0 > 2C + 10$; it will be enlarged once below by an amount depending only on C and K. Put

$$\sigma = \frac{1}{10} + \frac{\eta}{4}.$$

Thus $\sigma > 1/10$ and, for our parameters, $\sigma > 2\omega$. Apply the Heath–Brown identity with $K = 10$, for which $1/K < 2\sigma$, and use the finer-than-dyadic decomposition in [4, Equations (3.8)–(3.13)]. More explicitly, take

$$\Theta = 1 + (\log x)^{-C_0}$$

and a smooth partition of unity on the multiplicative scales $N \in \{\Theta^m : m \ge 0\}$. The cutoff functions ψ_N are supported on $[\Theta^{-1}N, \Theta N]$ and, for every fixed $r \ge 0$, satisfy

$$\sup_t N^r |\psi_N^{(r)}(t)| \ll_r (\log x)^{rC_0}. \tag{5.1}$$

The same bounds, with one additional fixed power of $\log x$, hold for the factor carrying the logarithm in the Heath–Brown identity. The implied constants are independent of the scale N.

For the term with j truncated Möbius factors, where $1 \le j \le K$, there are $2j \le 20$ localized factors. The support restriction to $[x, 2x]$ leaves at most

$$O\big((\log x)^{2j(C_0+1)}\big) \le O\big((\log x)^{20(C_0+1)}\big) \tag{5.2}$$

nonzero scale-tuples. The endpoint-collar function e created when the sharp cutoff is removed is supported on two intervals of combined length $O(x(\log x)^{-C_0})$ and satisfies $|e(n)| \ll \tau(n)^{O_K(1)}(\log x)^{O_K(1)}$, with exponents independent of C_0. Lemma 1.3 of [4], applied in each progression and to the averaged term, gives

$$\sum_{q \in \mathcal{R}(x)} |E_e(x; q, a)| \ll x(\log x)^{-C_0+O_K(1)} + x^{1/2+2\omega+o(1)}. \tag{5.3}$$

Indeed, the bound in one residue class is $((x(\log x)^{-C_0})/q + 1)x^{o(1)}$; summing uses $\sum_{q \le Q} q^{-1} \ll \log Q$ and $\sum_{q \le Q} \phi(q)^{-1} \ll \log Q$. This is the endpoint-collar estimate used in [4, Equations (3.10)–(3.13)]. Since $\omega < 1/20$, the second term in (5.3) has a fixed power saving. Enlarge C_0, if necessary, so that (5.3), the finitely many Heath–Brown coefficients, and the initial triangle inequalities contribute $O_C(x(\log x)^{-C})$.

We shall apply the two hypotheses of the lemma to each remaining localized convolution with the saving

$$C' = C + 20(C_0 + 1) + 2. \tag{5.4}$$

Such a saving is available because the hypotheses hold with arbitrary logarithmic saving. Summing (5.2) terms then leaves the required saving C.

We also verify that the distribution estimates are uniform over these scale-tuples. Every localized factor is one of $\psi_N \mu_{\le}$, ψ_N, or its logarithmic variant. Equations (5.1) and the divisor bound give a coefficient-sequence bound $(\log x)^{O_{C_0,K}(1)}$, uniformly in N. By [4, Lemma 3.4], a factor at any scale at least x^c, with fixed $c > 0$, has the Siegel–Walfisz property uniformly in the scale. The proof of that lemma also shows the required closure: if α and β have uniform coefficient-sequence bounds, β is Siegel–Walfisz, and its scale is at least x^c, then $\alpha \star \beta$ is Siegel–Walfisz, with a bound depending only on c, C_0, K and the requested saving. A group of at most twenty factors whose product scale is at least x^c contains a factor at scale at least $x^{c/20}$. That factor is Siegel–Walfisz by Lemma 3.4(iii), and iterating the closure statement over the other factors proves uniformly that the whole grouped convolution is Siegel–Walfisz. Multiplication by the single factor $\log N$ costs at most one power of $\log x$, which is covered by the two-power reserve in (5.4). These are precisely the uniformity conventions required for the Type II and Type III hypotheses.

It remains to locate the scales without losing an endpoint. For a nonzero scale-tuple $(N_1, \ldots, N_{2j})$, put

$$X_N = \prod_i N_i, \qquad t_i = \frac{\log N_i}{\log X_N}.$$

Then $t_i \geq 0$ and $\sum_i t_i = 1$ exactly. The support relations for the partition give

$$X_N = x^{1+O_K(1/\log x)}, \qquad \log_x N_i = t_i + O_K(1/\log x) \tag{5.5}$$

uniformly in the tuple. Since η is fixed, take x large enough that the error for every individual exponent and every sum of at most three exponents is smaller than $\eta/4$. Thus the decomposition reduces the von Mangoldt function on $[x, 2x]$, apart from the already bounded collar, to the uniformly controlled convolutions

$$\alpha_1 \star \cdots \star \alpha_s$$

whose normalized exponents are the t_i above. An individual factor with $t_i \geq 2\sigma$ is smooth by [4, Lemma 3.4(ii)]; equality causes no problem, because (5.5) and the common support enlargement by Θ give a scale comparable to $x^{2\sigma}$, which is the Vinogradov condition in that lemma.

Apply [4, Lemma 3.1] to the t_i. In its Type I/II alternative there is a partition into two groups, the smaller of which has normalized exponent θ satisfying

$$\frac{1}{2} - \sigma < \theta \leq \frac{1}{2}.$$

In view of (5.5), the exponent of the smaller group is greater than $1/2 - \sigma - \eta/4$. Both group scales are bounded below by a fixed positive power of x, so the uniform closure argument above shows that both grouped convolutions are Siegel–Walfisz. Moreover,

$$\frac{1}{2} - \sigma - \frac{\eta}{4} = \frac{2}{5} - \frac{\eta}{2} > \frac{2}{5} - \eta.$$

The other endpoint is at most $1/2 + \eta/4 < 3/5 + \eta$. Consequently this is a Type II convolution from the family in the statement.

In the Type III alternative there are three distinct smooth factors with exponents t_i, t_j, t_k satisfying

$$2\sigma \leq t_i, t_j, t_k \leq \frac{1}{2} - \sigma, \qquad t_i + t_j, t_i + t_k, t_j + t_k \geq \frac{1}{2} + \sigma.$$

The identities

$$2\sigma = \frac{1}{5} + \frac{\eta}{2}, \qquad \frac{1}{2} - \sigma = \frac{2}{5} - \frac{\eta}{4}, \qquad \frac{1}{2} + \sigma = \frac{3}{5} + \frac{\eta}{4}$$

place this convolution strictly inside the Type III ranges for $\xi_3 = 2/5$: the respective margins before the error in (5.5) are $3\eta/2$, $5\eta/4$, and $5\eta/4$. Hence all remain positive after an error of at most $\eta/4$. The three selected factors are smooth uniformly by the preceding paragraph. It is therefore covered by the second hypothesis.

It remains to treat the Type 0 alternative. Write the convolution as

$$F = \alpha \star \beta, \qquad \alpha(m) = \psi(m/N), \qquad N \geq x^{1/2+\sigma+o(1)}, \qquad M \asymp x/N,$$

where β is the complementary coefficient sequence at scale M and ψ has the smoothness bounds furnished by Lemma 3.4 of that paper. For primitive $a \pmod q$, terms with $(\ell, q) > 1$ vanish from both sides of the discrepancy. Put

$$S_\ell(a;q) = \sum_{\substack{x/\ell \leq m \leq 2x/\ell \\ m \equiv a\bar{\ell} \pmod q}} \alpha(m), \qquad\qquad S_\ell^*(q) = \sum_{\substack{x/\ell \leq m \leq 2x/\ell \\ (m,q)=1}} \alpha(m).$$

Then

$$E_F(x;q,a) = \sum_{(\ell,q)=1} \beta(\ell) \left(S_\ell(a;q) - \frac{S_\ell^*(q)}{\phi(q)} \right).$$

For fixed ℓ, the function

$$g_\ell(t) = \psi(t/N)\mathbf{1}_{[x/\ell, 2x/\ell]}(t)$$

has supremum and total variation bounded by $(\log x)^{O(1)}$. Euler summation in a residue class, and Möbius inversion for the coprimality condition, therefore bound the expression in parentheses by

$$\tau(q)(\log x)^{O(1)} = x^{o(1)}.$$

Since $\sum_\ell |\beta(\ell)| \ll M(\log x)^{O(1)}$ and $\#\mathcal{R}(x) \leq x^{1/2+2\omega}$, we obtain

$$\sum_{q \in \mathcal{R}(x)} |E_F(x;q,a)| \ll x^{1/2+2\omega} M(\log x)^{O(1)} \ll x^{1-\sigma+2\omega+o(1)}.$$

This is $O_C(x(\log x)^{-C})$ for every C, because $\sigma > 2\omega$. We have proved the required distribution estimate for Λ with the cutoff included.

Finally remove prime powers. On primes, $\Lambda(n) = \log n$, while the difference is supported on p^j with $j \geq 2$. Uniformly for primitive a, squarefreeness gives at most $2^{\#\{p:p|q\}}$ roots to $z^2 \equiv a \pmod q$. Hence the total progression contribution of prime squares, summed over $x^{1/2-2\tau} \leq q \leq x^{1/2+2\omega}$, is

$$\ll x^{1/2+2\omega+2\tau+o(1)}.$$

For $j \geq 3$, the crude bound of $O(x^{1/3})$ possible bases for each modulus gives

$$\ll x^{5/6+2\omega+o(1)}.$$

The averaged main terms contribute only $x^{1/2+o(1)}$, using $\sum_{q \leq Q} \phi(q)^{-1} \ll \log Q$. All three bounds are $O_C(x(\log x)^{-C})$ for every C. Dividing by $\log(3x)$ and using (2.10) proves the assertion for ρ_x. $\square$

Proposition 5.2. *The parameters* (4.1) *satisfy* (2.11).

Proof. Fix $\varepsilon_0 > 0$ and a required logarithmic saving. Choose $\tau > 0$ sufficiently small for the distribution estimates below, with $\tau \leq \tau_0$. The subfamily $q < x^{1/2-2\tau}$ is covered directly, for ρ_x, by the Bombieri–Vinogradov theorem for Λ, removal of prime powers, and division by $\log(3x)$. We therefore work on

$$\mathcal{Q}^+ = \{q \in \mathcal{Q}(x;\varepsilon_0) : q \geq x^{1/2-2\tau}\}.$$

The logarithms of all large factors have combined sum $M \leq 2B$. The smooth prime logarithms are less than δ, and the total logarithm of q is at least $1/2 - 2\tau$.

We verify the divisor conditions of [7, Lemmas 3, 4, 6, and 7]. All inequalities used for this purpose are strict. The numerical comparisons in this proof are rational comparisons, not floating-point tests; the exact checks are recorded in Appendix A.

Type IIa and Type IIb. Normalize the auxiliary scales so that $MN = x$; this changes M by only a bounded factor and preserves the coefficient-sequence and Siegel–Walfisz properties. Interchange the two factors when necessary, so that the smaller scale has exponent at most $1/2$, as in [7, Section 4.3, Case II]. Put

$$K = \frac{2}{5} + \frac{24}{5}\omega + \frac{7}{5}\delta = \frac{149677}{327680}. \tag{5.6}$$

For $\gamma \geq K + 2\eta$, set

$$\Delta_a(\gamma) = \frac{5\gamma - 2 - 24\omega}{7} - \eta.$$

One has $\Delta_a(\gamma) - \delta \geq 3\eta/7 > 0$, and

$$24\omega + 7\Delta_a(\gamma) - 5\gamma = -2 - 7\eta < -2,$$
$$8\omega + 3\Delta_a(\gamma) - \gamma = \frac{8\gamma - 6 - 16\omega}{7} - 3\eta < 0.$$

These are the scalar conditions of [7, Lemma 3]. The divisor interval has logarithmic endpoints $\gamma - 3\tau - \Delta_a(\gamma)$ and $\gamma - 3\tau$. Again include all large factors and fill with smooth primes. The upper endpoint exceeds $2B$, while

$$\frac{1}{2} - 2\tau - (\gamma - 3\tau - \Delta_a(\gamma)) \geq \frac{1}{14} - \frac{24}{7}\omega - \eta + \tau > 0.$$

For $G + 3\eta < \gamma < K + 2\eta$, put

$$\Delta_b(\gamma) = \frac{3\gamma - 1 - 24\omega}{7} - \eta.$$

Then $\Delta_b(\gamma) - \delta > 2\eta/7$, and

$$24\omega + 7\Delta_b(\gamma) - 3\gamma = -1 - 7\eta < -1,$$
$$8\omega + 3\Delta_b(\gamma) - \gamma = \frac{2\gamma - 3 - 16\omega}{7} - 3\eta < 0.$$

We apply [7, Lemma 4]. Choose u using only smooth primes, with logarithm in

$$\left(\frac{1}{2} - \gamma - 2\omega - 6\tau - \Delta_b(\gamma),\ \frac{1}{2} - \gamma - 2\omega - 6\tau\right).$$

The upper endpoint is positive, and the smooth mass exceeds even that upper endpoint: it is enough to check

$$G + 2\omega - 2B - 2\tau_0 > 0. \tag{5.7}$$

Next choose r, coprime to u, containing all large factors, with logarithm in

$$(\gamma - 3\tau - \Delta_b(\gamma),\ \gamma - 3\tau).$$

Its initial logarithm is below the upper endpoint, by $2B < G + 3\eta - 3\tau_0$. The remaining logarithm exceeds the lower endpoint, since

$$\log_x(q/u) > \gamma + 2\omega + 4\tau > \gamma - 3\tau - \Delta_b(\gamma).$$

Lemma 3.3 therefore produces both divisors.

Type IIc. It remains to consider $\gamma_- \le \gamma \le \gamma_+$. With $\delta_* = \delta + \zeta$, exact comparison gives

$$1 - 8\omega - 4\delta_* - 2\gamma_+ > 0.02352, \tag{5.8}$$

$$\gamma_- - 32\omega - 10\delta_* > 0.01110, \tag{5.9}$$

$$4\gamma_- - 1 - 48\omega - 16\delta_* > 0.00110. \tag{5.10}$$

Thus all three scalar hypotheses of [7, Lemma 6] hold throughout the interval. The required divisors are exactly those in Proposition 4.1. Notice that its argument includes $-\tau \le \omega_0 < 0$ as well as $\omega_0 \ge 0$.

Type III. Use the scale parameter $\gamma_3 = \xi_3 + \eta$ and put

$$\Delta_3 = \frac{1}{2} - \frac{7}{2}\omega - \frac{9}{8}\gamma_3 - \eta. \tag{5.11}$$

Then $\Delta_3 > \delta$ and

$$28\omega + 9\gamma_3 + 8\Delta_3 = 4 - 8\eta < 4.$$

The Type III scales in $\mathcal{H}(\xi_1, \xi_2, \xi_3)$ satisfy the scale conditions of [7, Lemma 7] with γ_3. Set

$$U_3 = \frac{1}{3} + \frac{4}{3}\Delta_3 - \frac{4}{3}\omega, \qquad L_3 = U_3 - \Delta_3. \tag{5.12}$$

That lemma requires a divisor with logarithm in (L_3, U_3). If $M < U_3$, place all large factors in the divisor. If $M \ge U_3$, exclude any one large factor. Its logarithm lies in $[\delta, b_1]$, so the logarithm of the remaining large factors is at most $2B - \delta < U_3$. In either case the available total logarithm exceeds L_3, since

$$U_3 - (2B - \delta) > 0.00307, \qquad \frac{1}{2} - 2\tau_0 - b_1 - L_3 > 0.01303. \tag{5.13}$$

Fill with smooth primes and apply Lemma 3.3.

Passage to primes. We have obtained the required distribution estimate on $\mathcal{Q}^+$ for every Type II and Type III convolution required in Lemma 5.1. The constants and auxiliary parameters can be chosen uniformly on the indicated ranges: the scalar inequalities have positive margins, and the cutoff ranges have the fixed tolerance η. The bound (2.9) places every modulus within $x^{1/2+2\omega}$. Since $\omega = 239/32768 < 1/20$, Lemma 5.1 proves (2.11) on $\mathcal{Q}^+$. Combining it with Bombieri–Vinogradov on the smaller moduli completes the proof. □

6. A finite sieve function

We next give an exact discretization for evaluating (2.4) and a lower bound for (2.5). The formulas in this section apply to any parameters satisfying (2.1) that are integral multiples of a mesh $d > 0$, with the index ranges below nonempty. For definiteness, assume $1 \le D < C_1 \le N+1$ and $0 \le M_0 \le N$.

Write

$$\begin{gathered} D = \delta/d, \quad C_r = B_r/d, \quad T = (A+\varepsilon)/d, \quad L = (A-\varepsilon)/d, \\ N = T - k, \qquad M_0 = L - (k-1). \end{gathered} \tag{6.1}$$

For $a \ge 0$ let $Q_a = [ad, (a+1)d)$. A box $Q_{a_1} \times \cdots \times Q_{a_k}$ is retained if

$$s = \sum_i a_i \le N, \qquad z + r \le C_r, \quad r = \#\{i : a_i \ge D\}, \quad z = \sum_{a_i \ge D} a_i, \tag{6.2}$$

where the second condition is omitted for $r = 0$. Every retained box lies in $\mathcal{T}_k$ up to a null set. The use of upper endpoints in (6.2) is important: a retained box is not merely a box whose centre lies in the support.

Fix positive arrays $g_\ell(a)$ and real arrays $h_\ell(s)$, with $0 \le \ell < R$ and $0 \le a, s \le N$. On a retained box define

$$F(t_1, \dots, t_k) = \sum_{\ell=0}^{R-1} h_\ell(s) \prod_{i=1}^{k} g_\ell(a_i), \tag{6.3}$$

and set $F = 0$ elsewhere. This function is symmetric, bounded, and square-integrable. No sign condition is imposed on the radial arrays.

6.1. **Convolution coefficients.** For $0 \le \ell, m < R$, put

$$q_{\ell m}(a) = d\, g_\ell(a) g_m(a).$$

Let $q^{\mathrm{s}}_{\ell m}$ be its restriction to $0 \le a < D$, extended by zero, and let $q^{\mathrm{l}}_{\ell m}$ be its restriction to $D \le a < C_1$. The latter truncation is legitimate by (2.3). Define

$$P^{\ell m}_j = (q^{\mathrm{s}}_{\ell m})^{*j}, \qquad P^{\ell m}_0(s) = \mathbf{1}_{s=0}, \tag{6.4}$$

$$Q^{\ell m}_0(z) = \mathbf{1}_{z=0}, \qquad Q^{\ell m}_r(z) = \mathbf{1}_{z \le C_r - r} (Q^{\ell m}_{r-1} * q^{\mathrm{l}}_{\ell m})(z). \tag{6.5}$$

Here $*$ denotes convolution of sequences indexed by the nonnegative integers. In every calculation coefficients beyond the largest needed index are discarded.

The truncation after each convolution in (6.5) loses no term that could contribute at a later stage. Indeed, for $r \ge 2$, if r large indices have sum $z \le C_r - r$, deleting an index at least D leaves sum at most

$$C_r - r - D \le C_{r-1} - r < C_{r-1} - (r-1).$$

This uses $C_r - C_{r-1} \le D$. Iterating the argument reduces to $r = 1$, where the cap is $C_1 - 1$. It also shows directly that the individual large indices are less than C_1.

Proposition 6.1. *For the function* (6.3),

$$I(F) = \sum_{\ell,m=0}^{R-1} \sum_{s=0}^{N} h_\ell(s) h_m(s) \sum_{r=0}^{k} \binom{k}{r} (P^{\ell m}_{k-r} * Q^{\ell m}_r)(s). \tag{6.6}$$

Only values $r \le \min(k, \lfloor B/\delta \rfloor)$ can contribute when $B_r \le B$.

Proof. Expand the square in (6.3). Select the r large coordinates in $\binom{k}{r}$ ways. Their index sum is recorded by $Q^{\ell m}_r$, and the sum of the remaining indices by $P^{\ell m}_{k-r}$. Each factor $q_{\ell m}$ includes one factor d, so the product already includes the volume d^k of the box. Summing over all retained boxes proves the formula. □

6.2. **The box numerator.** In the $(k-1)$ common coordinates of (2.5), retain only boxes whose upper-endpoint sum is at most $A - \varepsilon$. Their index sum is therefore at most M_0. Let $J_{\mathrm{box}}(F)$ be the integral in (2.5) over this union of common-coordinate boxes, with the inner fiber integral left unchanged. Since the integrand is a square,

$$0 \le J_{\mathrm{box}}(F) \le J(F). \tag{6.7}$$

For a common-coordinate box with $s \le M_0$, large-coordinate count r, and large index sum z, first impose the inherited cap $z + r \le C_r$ when $r > 0$. A box failing this cap has an empty fiber. For a box satisfying it, the allowed last-coordinate indices are exactly $0 \le a \le u(s, z, r)$, where

$$u(s, z, r) = \min\big(N - s,\ \max(D - 1, C_{r+1} - r - 1 - z)\big). \tag{6.8}$$

This follows by considering separately a small last coordinate and a large last coordinate. Define the fiber prefixes

$$H_\ell(s, u) = \sum_{a=0}^{u} g_\ell(a) h_\ell(s + a). \tag{6.9}$$

Proposition 6.2. *For F in* (6.3),

$$J_{\mathrm{box}}(F) = d^2 \sum_{\ell,m=0}^{R-1} \sum_{s=0}^{M_0} \sum_{r=0}^{k-1} \binom{k-1}{r} \sum_{z\ge 0} Q_r^{\ell m}(z) P_{k-1-r}^{\ell m}(s-z) \cdot H_\ell(s,u(s,z,r)) H_m(s,u(s,z,r)). \quad (6.10)$$

Terms with indices outside the arrays are zero.

Proof. The inner integral on a fixed common-coordinate box is

$$d \sum_{\ell=0}^{R-1} \left(\prod_{i=1}^{k-1} g_\ell(a_i) \right) H_\ell(s,u(s,z,r)).$$

Square this expression and group the common-coordinate boxes by s, r, z. Their product weights and volumes give $\binom{k-1}{r} Q_r^{\ell m}(z) P_{k-1-r}^{\ell m}(s-z)$. The two remaining factors of d come from the two fiber integrals. ☐

6.3. **The coefficient arrays.** For (4.1), choose

$$d = \frac{1}{65536}, \qquad R = 3. \quad (6.11)$$

The integer data in (6.1) are

$$\begin{gathered} D = 1019, \quad T = 17419, \quad L = 16305, \quad N = 17374, \quad M_0 = 16261, \\ C_1 = 10171, \quad C_2 = 10378, \quad C_3 = 11321, \quad C_r = 12082 \ (r \ge 4). \end{gathered} \quad (6.12)$$

At most eleven large coordinates occur. The six arrays $g_0, g_1, g_2, h_0, h_1, h_2$, each of length 17375, are specified in Appendix A. Every entry is an exact dyadic rational; together with (6.2) and (6.3), these arrays define F exactly.

The arrays were found by a variational search in the family (6.3). The coordinate profiles were initialized from

$$\left(\frac{c}{c + (a+\frac{1}{2})d} \right)^{p_\ell}, \qquad c = 0.0014, \quad (p_0,p_1,p_2) = (0.90, 0.96, 1.02),$$

with an independent normalization for each profile. The radial arrays were optimized on a coarser mesh and interpolated to (6.11). This describes the search only. The definition used in the proof is the stored dyadic array, not the displayed real-power formula or the output of an eigenvalue solver.

7. The certificate and the prime-gap bound

Proposition 7.1 (Finite certificate)**.** *The function specified by* (6.3), (6.12), *and the coefficient arrays in Appendix A has $I(F) > 0$ and satisfies*

$$\frac{6037239562351545}{6036062711329456} \le \frac{45 J_{\mathrm{box}}(F)}{I(F)} \le \frac{7244687476684354 5}{72432752528865376}. \quad (7.1)$$

In particular,

$$1.0001949699 < \frac{45 J_{\mathrm{box}}(F)}{I(F)} < 1.0001949704. \quad (7.2)$$

Proof. Evaluate (6.6) and (6.10) by directed rounding, with the input entries interpreted as exact dyadic rationals. The interval computation performs the convolutions (6.4)–(6.5) directly and evaluates the prefix sums (6.9). It returns lower and upper bounds for each pair (ℓ, m) with $\ell \le m$. Off-diagonal contributions are included with multiplicity two.

The polynomial coefficients are nonnegative. Their lower and upper bounds are therefore obtained by downward and upward rounding, respectively, at every operation. The factors involving h_ℓ may have either sign; these are enclosed by interval multiplication and addition. The resulting hexadecimal endpoints are converted to exact rational numbers before the six pair contributions are combined. In particular, the exact aggregate enclosures are

$$\frac{2263523516527043}{2251799813685248} \le I(F) \le \frac{1131761758374273}{1125899906842624},$$
$$\frac{402482637490103}{18014398509481984} \le J_{\mathrm{box}}(F) \le \frac{1609930550374301}{72057594037927936}.$$

The norm has a strictly positive lower bound. Dividing the lower numerator by the upper denominator, and conversely for the upper bound, gives (7.1). The arithmetic procedure and complete endpoint data are specified in Appendix A. □

Proof of Theorem 1.2. Proposition 5.2 verifies the distribution hypothesis of Proposition 2.1. The function F is symmetric and belongs to $L^2(\mathcal{T}_{45})$. By (6.7) and Proposition 7.1,

$$45J(F) \ge 45J_{\mathrm{box}}(F) > I(F) > 0.$$

Proposition 2.1 proves the assertion. □

Proof of Theorem 1.1. Consider the following set $\mathcal{A}$:

$$\begin{aligned}\{&0, 2, 12, 14, 24, 26, 30, 32, 36, 50, 54, 56, 60, 66, 72, 74, 80, 84,\\ &92, 96, 102, 110, 114, 116, 122, 126, 134, 140, 144, 150, 156, 162,\\ &164, 170, 176, 180, 182, 186, 192, 194, 200, 204, 206, 210, 212\}.\end{aligned} \tag{7.3}$$

It has 45 distinct elements and diameter 212. The table below gives one residue class omitted by $\mathcal{A}$ for each prime at most 45.

p	2	3	5	7	11	13	17	19	23	29	31	37	41	43
Omitted	1	1	3	6	9	3	1	6	17	13	28	31	8	25

For $p > 45$, a 45-element set cannot occupy all residue classes. Thus $\mathcal{A}$ is admissible. By Theorem 1.2, infinitely many of its translates contain two primes. Between any such pair there is a consecutive-prime gap at most 212. These pairs occur arbitrarily far out, proving $H_1 \le 212$. □

Acknowledgments

Siddhartha thanks Paras Chopra, founder of Lossfunk, for introducing him to the general research direction of using AI systems to attack mathematical problems. He also thanks Lossfunk, a Bangalore-based research lab, for providing the research workspace, environment, and GPT access that enabled this project. Siddhartha also thanks Julia Stadlmann, whose work [7] this manuscript heavily builds upon.

Appendix A. Finite certificate

The arrays in (6.3) are specified by `ancillary/witness.hex`. Its first line contains $(k, R, N+1, D, d^{-1}, T, L, r_{\max})$, and its second line contains $(C_0, \ldots, C_{r_{\max}+1})$, where $k = 45$, $C_0 = 0$, and $r_{\max} = 11$; the remaining parameter values are given in (6.11)–(6.12). For each $a = 0, \ldots, N$, the next line lists $g_0(a), g_1(a), g_2(a), h_0(a), h_1(a), h_2(a)$ as exact hexadecimal dyadic rationals. The SHA–256 digest, with the two lines concatenated, is

```
4b82ce9794faeb80d1766dddc2884d6a
19734490baf15a4200bb15bdfd5af718
```

The verifier `ancillary/certify.cpp` evaluates (6.6) and (6.10) using IEEE 754 binary64 arithmetic with gradual underflow and correctly rounded addition, multiplication, and division. The input dyadics and all required binomial coefficients are represented exactly. Each operation is rounded outwards; signed products use the minimum and maximum of the four endpoint products. Induction therefore gives an enclosure of every intermediate quantity. The six pair enclosures are combined with exact rational arithmetic, giving (7.1). The computation assumes that the implementation honors the rounding modes and compiler options

```
-O3 -std=c++17 -fopenmp -frounding-math
-ffp-contract=off -fno-fast-math
```

With Python 3 and a C++17 compiler supporting this arithmetic model and OpenMP, the certificate is reproduced by

```
python3 ancillary/run_certificate.py --threads 4
```

This checks the witness digest, the rational support inequalities, and tuple admissibility, then compiles the verifier and evaluates the finite sums. The recorded pair endpoints are in `ancillary/reference_4threads.json`.

The six margins in (4.6)–(4.7) exceed, in their displayed order,

$$(0.0001576,\ 0.0000751,\ 0.0001434,\ 0.02725,\ 0.0001891,\ 0.0001869).$$

These terminating decimals are strict rational lower bounds. All parameter comparisons in Section 5 are checked by `ancillary/verify_support.py`; their exact margins are recorded in `ancillary/support_output.json`.

Appendix B. Research provenance and release status

B.1. **Generation and human involvement.** This manuscript was generated in the ChatGPT web interface by OpenAI GPT-5.6 Sol (model identifier `gpt-5.6-sol`), using Pro reasoning effort. Siddhartha Mahajan, Research Fellow at Lossfunk, supplied the bounded-prime-gaps problem, directed GPT-5.6 Sol to the relevant source materials, and gave brief, high-level instructions, including the instruction to seek the strongest reduction it could support. GPT-5.6 Sol generated the principal mathematical argument, parameter search, manuscript text, certificate data, and verification software. Codex was used afterward to inspect the release package, rerun its computational checks, and assist with verification and preparation of the final text.

There has been no human verification of this work to date.

The model is named as the author of this manuscript to reflect how the work was produced.

B.2. **Public archive.** The manuscript source, ancillary code, exact witness, reference outputs, checksums, and a human-readable provenance record are available at

`https://github.com/Siddhartha-Mahajan/prime-gap-212`.

References

[1] R. C. Baker and A. J. Irving, *Bounded intervals containing many primes*, Math. Z. **286** (2017), 821–841.
[2] D. A. Goldston, J. Pintz, and C. Y. Yıldırım, *Primes in tuples I*, Ann. of Math. (2) **170** (2009), 819–862.
[3] J. Maynard, *Small gaps between primes*, Ann. of Math. (2) **181** (2015), 383–413.
[4] D. H. J. Polymath, *New equidistribution estimates of Zhang type*, Algebra Number Theory **8** (2014), no. 9, 2067–2199.
[5] D. H. J. Polymath, *Variants of the Selberg sieve, and bounded intervals containing many primes*, Res. Math. Sci. **1** (2014), 1–83.
[6] J. Stadlmann, *On primes in arithmetic progressions and bounded gaps between many primes*, Adv. Math. **468** (2025), 110190.
[7] J. Stadlmann, *Bounded gaps between primes*, arXiv:2608.31126v1 [math.NT], 2026.
[8] Y. Zhang, *Bounded gaps between primes*, Ann. of Math. (2) **179** (2014), no. 3, 1121–1174.